

Financial Market Reactions to Sustainability and Climate Policy Events

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Abstract

This study examines whether sustainability-related political developments generate measurable financial-market responses when they arrive as unexpected information shocks rather than gradual policy signals. Using an event-study framework, the analysis focuses on the July 2022 “Manchin Surprise,” when Senator Joe Manchin unexpectedly announced support for major U.S. climate legislation, sharply increasing the perceived feasibility of sustainability policy. By comparing market behavior before and after this event, and contrasting sustainability-aligned firms with a matched control group, the study evaluates whether sustainability has emerged as an identifiable factor in asset pricing when expectations about policy credibility shift.

The empirical approach combines cumulative abnormal return (CAR) analysis with Difference-in-Differences (DiD) regression framework. Expected returns are estimated using a market model benchmarked to the S&P 500, and abnormal returns are aggregated across a four-week pre- and post-event window. The sustainability-focused sample is drawn from Barron's 100 ESG list, while control firms are matched by sector and market capitalization. This structure isolates firm-specific responses to the political event from broader market movements and macroeconomic influences.

Results show a clear and persistent divergence in market behavior following the Manchin Surprise. Sustainability-aligned firms experience a statistically significant increase in cumulative abnormal returns in the post-event period, while matched control firms remain close to zero. Confidence intervals around mean CARs for ESG firms increasingly exclude zero after the event, indicating a systematic revaluation rather than short-term noise. In contrast, applying the same methodology to the formal passage of the Inflation Reduction Act yields weaker and less distinct market responses, consistent with much of its content having been priced in advance.

Regression results reinforce the event-study findings. The interaction between the post-event indicator and sustainability classification is positive and statistically significant, indicating that ESG-aligned firms experienced incremental gains beyond general market adjustments. Taken together, the evidence supports rejection of the null hypothesis and suggests that sustainability-related political developments can generate economically meaningful market effects when they resolve uncertainty and alter beliefs about policy credibility. The findings imply that sustainability now functions as a market-relevant information channel, shaping investor expectations and asset prices when political events unexpectedly shift the likelihood of long-term climate and energy policy implementation.

❖ Introduction

Over the past decade, financial markets have increasingly responded to sustainability not as a slow and continuous trend, but as a series of identifiable policy and political events. When governments signal changes in climate policy, energy regulation, or environmental commitments, markets quickly reassess expectations about future costs, profits, and investment opportunities. These reassessments tend to occur around specific moments, rather than gradually over time, making sustainability particularly well suited for time-based market analysis.

Sustainability-related policies are often debated, revised, or delayed, which creates uncertainty about whether stated goals will translate into real economic change. Markets therefore react most strongly when new information alters the perceived credibility of policy direction. Unexpected political developments can shift expectations sharply, even if formal legislation remains unchanged. In this sense, sustainability becomes relevant to markets not only through long-term objectives, but through discrete events that clarify or disrupt anticipated policy outcomes.

This event-driven perspective highlights the importance of timing. Market prices incorporate information as it arrives, and reactions around key dates reveal whether sustainability is viewed as economically meaningful or largely symbolic. Sudden changes in political alignment, voting behavior, or legislative feasibility can produce immediate market responses as investors adjust beliefs about the likelihood of future regulation and public investment.

The analysis that follows adopts this time-event approach to examine whether sustainability-related political developments generate measurable market effects. By isolating a clearly defined event and comparing market behavior before and after its occurrence, the framework evaluates whether sustainability functions as an identifiable factor in asset pricing when expectations shift.

❖ Background and Theoretical Motivation

The growing connection between sustainability and macroeconomic outcomes is increasingly evident in both aggregate economic data and financial-market behavior. Climate-related shocks—such as extreme weather events, energy supply disruptions, and environmental disasters, now exert measurable effects on output, inflation, and productivity across countries. These shocks influence not only short-term fluctuations but also long-run growth prospects by altering capital accumulation, labor productivity, and sectoral composition. As a result, sustainability has moved from a peripheral concern to a central factor shaping macroeconomic expectations.

Policy responses to these challenges play a critical role in translating environmental risks into market-relevant signals. Emissions targets, carbon pricing mechanisms, clean-energy subsidies, and industrial-policy initiatives directly affect firms' cost structures and investment incentives. Evidence from comparative studies on the coexistence of sustainability and growth shows that emissions reductions and economic expansion can occur simultaneously, particularly in economies that invest in energy efficiency, renewable infrastructure, and innovation (García-Monleón, F., González-Rodrigo, E., & Bordonado-Bermejo, M.-J. 2024). These outcomes tend to follow major policy commitments, suggesting that markets respond not merely to environmental goals themselves, but to the credibility and enforceability of the policies that support them.

Financial markets provide further confirmation of this mechanism. Firm-level research examining market reactions to major climate agreements demonstrates that companies with exposure to clean energy, emissions-reduction technologies, or environmentally aligned revenue streams often experience positive abnormal returns around key policy events. These reactions highlight an important insight: investors respond most strongly when sustainability initiatives signal durable shifts in future demand, regulation, or competitive advantage. In this sense, sustainability policy functions as an information event that reshapes expectations about long-term profitability rather than as a simple ethical preference.

Similar dynamics are visible beyond landmark global agreements. Renewable-energy firms, electric-vehicle producers, and companies tied to low-carbon infrastructure frequently experience valuation changes following regulatory announcements, subsidy expansions, or large public investment

programs. In contrast, firms in carbon-intensive sectors often face persistent downward pressure as markets anticipate higher compliance costs, tighter regulation, and the risk of stranded assets. These asymmetric responses suggest that sustainability has become a key axis along which capital is reallocated across sectors.

Green finance reinforces this pattern by channeling investment toward projects that reduce exposure to climate and energy risks. The expansion of green bonds, sustainability-linked loans, and climate-focused investment funds has sometimes generated uneven short-term market reactions, reflecting uncertainty about standards and verification (Kruse, T., Mohnen, M., & Sato, M. 2024). Over time, however, these instruments help stabilize markets by funding infrastructure that lowers vulnerability to energy price shocks and geopolitical disruptions. Observing market behavior before and after the introduction or expansion of such financial tools helps isolate the economic value of sustainability-oriented capital flows.

The theoretical foundation for analyzing these dynamics closely follows the event-study logic commonly used in financial economics. That framework emphasizes that markets react differently depending on whether an event alters long-term expectations or merely represents a temporary disturbance. Events that trigger strategic reassessment, such as credible shifts in government priorities or binding policy commitments, tend to generate larger and more persistent abnormal returns. Applied to sustainability, climate legislation or large-scale green investment programs should therefore elicit stronger market responses when they convincingly signal lasting changes in regulation, technology adoption, or capital allocation.

Historical financial research supports this interpretation. Early work by scholars such as White and Hamilton shows that markets consistently reward improvements in information quality, transparency, and productive efficiency. Modern sustainability disclosures, including standardized climate reporting and enhanced environmental risk metrics, function in much the same way. When these frameworks are introduced or strengthened, they create clear information events that allow investors to distinguish firms that are well positioned for a low-carbon transition from those that are not. Improved information reduces uncertainty,

enhances pricing efficiency, and reallocates capital toward firms with stronger long-term prospects.

Empirical studies examining major climate agreements and policy announcements reinforce this conclusion. Abnormal returns tend to cluster around event dates, particularly for firms directly exposed to clean energy, emissions regulation, or environmental innovation. These effects are strongest when announcements signal durable policy commitments rather than symbolic or contested gestures. When credibility is weak or political follow-through is uncertain, market reactions are typically smaller and short-lived, underscoring the importance of belief revision rather than policy rhetoric.

Event-study theory provides a clear explanation for these patterns. Markets respond most powerfully to events that resolve uncertainty and force investors to update expectations about the future regulatory and economic environment. Applying this framework to sustainability therefore requires careful event selection, focusing on moments when the perceived likelihood of policy success changes unexpectedly. Political breakthroughs, legislative turning points, or credible implementation signals offer a clearer lens through which to assess whether sustainability has become a structural driver of market behavior, one that consistently shapes returns, volatility, and valuations across diverse markets and economic conditions.

❖ Hypothesis Development

To guide the empirical analysis, this study establishes a clear and testable pair of hypotheses grounded in a time-period, event-study framework. The focus is not on whether sustainability is inherently beneficial or harmful, but on whether sustainability-related actions produce any statistically detectable market response when they occur.

Hypothesis:

- H_0 = no measurable change at all around the event window, and
- H_1 = some measurable change (of any sign or magnitude)

The null hypothesis (H_0) states that sustainability-oriented firms, policies, and investments produce no measurable change in market behavior during the event windows examined. Under this hypothesis, asset returns, volatility, and valuation metrics around sustainability-related events, such as climate legislation, major green-finance initiatives, or policy agreements, are statistically indistinguishable from normal market fluctuations. Any observed movements would reflect background market noise rather than a response to sustainability itself. In this case, sustainability actions would not meaningfully alter aggregate market performance or economic outcomes across the markets reviewed, implying that broader forces, such as monetary policy shifts, interest-rate cycles, geopolitical risk, commodity-price shocks, or non-sustainability-related technological change, remain the dominant drivers of market and economic dynamics.

The alternative hypothesis (H_1) posits that sustainability-related events do generate measurable changes in economic or financial-market outcomes during the event window. These changes may be positive, negative, or mixed, and need not be uniform across markets or sectors. Under this hypothesis, sustainability actions trigger detectable responses such as abnormal returns, changes in volatility, valuation adjustments, or shifts in capital allocation. This may appear through clean-technology sector outperformance, market revaluations following climate-policy announcements, sustained valuation premiums for firms with strong green-revenue exposure, or observable capital flows linked to green bonds and sustainability-linked financing. Even modest but statistically significant reactions would indicate that sustainability has become an identifiable factor influencing market behavior, consistent with recent literature linking green innovation, policy credibility, investor expectations, and long-run economic performance.

The distinction between these hypotheses reflects the reality that sustainability does not operate in isolation. Market reactions to sustainability events may be small relative to those generated by major macroeconomic forces. Central-bank interest-rate decisions, for example, often produce immediate and powerful market movements that can overshadow the effects of climate policy announcements. Similarly, geopolitical conflicts, supply-chain disruptions, commodity-price shocks, or major technological breakthroughs can dominate short-term market behavior. Recognizing these competing influences is central to the empirical

design. By focusing on time-specific event windows and controlling for broader market movements, the analysis seeks to isolate whether sustainability-related actions produce any systematic deviation from expected market behavior, or whether their effects remain indistinguishable from noise.

❖ Event Study Tests of Market Response

To evaluate the hypotheses, this study applies an event study approach that focuses on cumulative abnormal return trends around sustainability related events. The goal is to identify whether markets behave differently during periods when new sustainability information becomes available, such as the announcement of climate legislation, major green investment programs, or international policy commitments. By examining market behavior before and after these events, the analysis can determine whether sustainability actions generate responses that differ from what would normally be expected based on overall market movements.

Cumulative abnormal returns serve as the primary test of whether markets react to sustainability events. Abnormal returns are calculated by comparing actual asset returns to expected returns based on broader market performance. If sustainability has no effect, abnormal returns should remain close to zero throughout the event window. Under the null hypothesis, any small movements in returns would reflect ordinary market noise rather than a response to the event itself. Under the alternative hypothesis, cumulative abnormal returns may diverge from zero following the event, indicating that investors reassess firm value, sector prospects, or market expectations in response to sustainability related information. These responses may be positive or negative and may persist beyond the immediate event date if the event signals a lasting change in policy direction or economic structure.

In addition to examining cumulative abnormal returns, the analysis uses a Difference-in-Differences (DiD) to evaluate whether sustainability-related events produce different market responses for sustainability-aligned firms relative to a broad market control group. Rather than focusing on volatility as a separate outcome, this approach directly tests whether the magnitude and direction of market reactions differ between firms expected to be affected by the event and those representing general market conditions. This comparison provides a clearer

assessment of whether sustainability influences market behavior beyond overall market movements.

The regression model compares outcomes for the sustainability-focused sample and the control group across the event window. Cumulative abnormal returns are regressed on an indicator for the event period, an indicator for membership in the sustainability-focused group, and an interaction term capturing their joint effect. The interaction term isolates whether sustainability-aligned firms experience a different response to the event than the market as a whole. If sustainability is not a relevant factor, the interaction effect should be statistically insignificant, indicating that both groups move similarly during the event window.

This framework allows the analysis to confirm whether observed abnormal returns are specific to sustainability-aligned firms or simply reflect broader market reactions. By explicitly modeling the difference between the test group and the control group, the regression separates sustainability-driven effects from movements caused by interest rate changes, macroeconomic announcements, or other external shocks that affect all firms simultaneously. A statistically significant interaction term provides evidence that sustainability alters how firms respond to policy or political events.

Together, the cumulative abnormal return analysis and the Difference-in-Differences (DiD) regression offer a rigorous test of sustainability's role in market behavior. The event study identifies whether markets react around key sustainability-related events, while the regression framework verifies whether those reactions are stronger or weaker for sustainability-aligned firms relative to the broader market. Consistent differences between the test and control groups would support the view that sustainability has become an identifiable factor influencing how investors assess risk, revalue assets, and allocate capital over time.

❖ Methodology

This study uses an event study methodology to measure how financial markets respond to sustainability related events over time. The central idea is to compare

actual market behavior around an event to what would normally be expected if the event had not occurred. Any difference between actual and expected outcomes is interpreted as the market response to the sustainability action. This method allows the analysis to isolate sustainability driven effects from broader market movements.

Estimation of Expected Returns

Let $R_{i,t}$ represent the actual return of asset (i) on day (t). Expected returns are estimated using a market model based on historical data prior to the event. For each asset, expected returns are calculated as

$$E(R_{i,t}) = \alpha_i + \beta_i R_{m,t}$$

where $R_{m,t}$ is the return on the relevant market index on day (t), and (α_i) and (β_i) are estimated using ordinary least squares over an estimation window that precedes the event. The estimation window is chosen to be sufficiently long so that parameter estimates reflect normal market conditions and are not influenced by the event itself.

Abnormal Returns and Cumulative Abnormal Returns

Abnormal returns measure the portion of an asset's return that cannot be explained by overall market movements. The abnormal return for asset (i) on day (t) is defined as

$$AR_{i,t} = R_{i,t} - E(R_{i,t})$$

If sustainability related events have no effect on markets, abnormal returns should be close to zero throughout the event window. To evaluate the total market response over time, abnormal returns are aggregated to form cumulative abnormal returns. For an event window spanning from day (t_1) to day (t_2), cumulative abnormal returns are calculated as

$$CAR_i(t_1, t_2) = \sum_{t=t_1}^{t_2} AR_{i,t}$$

Cumulative abnormal returns provide a clear measure of whether markets experience systematic revaluation following a sustainability event. Persistent deviations from zero suggest that the event altered investor expectations in a meaningful way.

Regression Analysis of Market Variation

In addition to cumulative abnormal returns, this study employs a Difference-in-Differences (DiD) regression framework to examine whether sustainability-related events are associated with systematic changes in market behavior. Rather than modeling volatility directly, the regression framework evaluates variation in market outcomes while controlling for firm characteristics and event timing. This approach allows the analysis to isolate whether sustainability-related actions coincide with measurable changes in performance during the event window.

The empirical specification used in this analysis is given by:

$$CAR_i = \alpha + \beta_1 Post_t + \beta_2 ESG_i + \beta_3 (Post_t \times ESG_i) + \varepsilon_{i,t}$$

where $CAR_{i,t}$ denotes the cumulative abnormal return for firm i over the event window. The indicator variable $Post_t$ equals one for observations occurring after the sustainability-related event and zero otherwise. The variable ESG_i identifies firms classified as sustainability-focused, while the interaction term $Post_t \times ESG_i$ captures the product of ESG and Post boolean value. The error term $\varepsilon_{i,t}$ reflects unexplained variation in returns.

Under the null hypothesis, the coefficient on the interaction term, β_3 , is not statistically different from zero. This outcome would indicate that sustainability-related events do not lead to abnormal performance for ESG firms beyond general market movements following the event. In this case, observed changes in cumulative abnormal returns would reflect background market variation rather than a sustainability-driven response. Under the alternative hypothesis, β_3

differs from zero, suggesting that sustainability-focused firms experience systematically different market reactions following the event.

This regression framework can be extended across different firm groupings, sectors, and market segments. Separate estimations may be conducted for sustainability-focused firms, carbon-intensive firms, and broad market benchmarks. Comparing coefficients across these specifications allows the analysis to assess whether sustainability effects are concentrated within specific segments or whether they reflect broader market-wide dynamics.

By combining cumulative abnormal returns with this interaction-based regression analysis, the methodology provides a robust test of whether sustainability has emerged as an identifiable factor in financial markets. The event study captures immediate market reactions, while the regression framework evaluates whether those reactions differ systematically across firm types after accounting for event timing. Together, these methods offer a clear assessment of whether sustainability-related actions generate measurable market effects or remain indistinguishable from normal market behavior.

❖ **Event Selection, Timeframe, and Sample Construction**

With the testing framework and hypotheses established, the analysis turns to the definition of the event, the observation windows, and the construction of the test and control samples. The focal event is the **Manchin Surprise**, a sudden political development in mid-July 2022 in which Senator Joe Manchin unexpectedly announced support for a climate and energy package after previously signaling opposition. This announcement materially altered market expectations regarding the likelihood of major U.S. sustainability legislation. Unlike the formal passage of the Inflation Reduction Act, whose key parameters were increasingly anticipated, the Manchin Surprise represents a clear and concentrated information shock that is well suited for an event-study framework (Colman, Z., Siegel, J., & Tamborrino, K. 2022).

The use of the Manchin Surprise allows the analysis to isolate the impact of new information rather than the confirmation of already known policy outcomes. By focusing on the moment when legislative feasibility shifted unexpectedly, the

design captures how markets respond when beliefs about sustainability policy credibility change. The later passage of the Inflation Reduction Act provides context but is not treated as the primary event, as much of its content had already been priced into markets by that point.

The primary event window consists of four weeks prior to the Manchin Surprise and four weeks following it. This window captures both anticipatory market behavior as expectations evolve and adjustment dynamics as investors incorporate the new information into asset prices.

A longer control period is also included to establish baseline market behavior, allowing us to calculate alpha and beta per security used. This control period spans one full year prior to the start of the pre-event window. Data from this period are used to estimate expected returns and typical market relationships under normal conditions, before the sustainability-related political shift became relevant. This approach helps distinguish event-driven responses from broader market trends such as interest rate changes, inflation pressures, or general economic conditions.

The empirical analysis focuses on **Mean Cumulative Abnormal Returns (CAR)** and **Mean Cumulative Average Abnormal Returns (CAAR)**. The test group is drawn from the **Barron's 100 ESG list**, sourced from Barron's 2022 report, which relies on data provided by **Calvert Research and Management**. Calvert's selection process evaluates the one thousand largest publicly held U.S. companies, measured by market capitalization and excluding real estate firms, to identify those with strong environmental, social, and governance performance (Foster, L 2022 and Calvert Research and Management.). After cross-referencing Barron's list with available financial data, the final sample consists of approximately ninety-seven companies. This list is cross-referenced with available data tickers to eventually produce a final sample size of ~93 companies.

Using this ESG-focused sample ensures that the analysis concentrates on firms most likely to be affected by changes in sustainability policy expectations. These companies have greater exposure to potential shifts in clean energy incentives, regulatory standards, and climate-related investment flows. As a result, any market response to the Manchin Surprise should be more pronounced within this group.

To evaluate whether observed responses are specific to sustainability-aligned firms rather than reflective of broader market movements, a matched control group is constructed. Control firms are selected by attempting to match each ESG firm with a non-ESG firm of similar sector classification and comparable market capitalization. Matching on sector helps control for industry-specific shocks, while matching on firm size reduces differences related to scale, liquidity, and investor attention.

The **S&P 500** is used as the baseline market benchmark for estimating expected returns. Abnormal returns are calculated relative to index performance, allowing deviations to reflect firm-specific responses rather than general market trends. CAAR are then computed across the ESG sample and compared with those of the matched control group. This structure allows for a direct comparison of how sustainability-aligned firms and similar non-ESG firms respond to the same political information shock.

Taken together, this design integrates a clean event definition, a well-targeted sustainability-focused sample, and a carefully matched control group. By focusing on Cumulative Average Abnormal Returns around the Manchin Surprise, the analysis provides a clear test of whether sustainability-related information alters market behavior, or whether observed movements remain indistinguishable from normal market variation driven by broader economic forces.

❖ **Study-In action and Results**

The analysis begins by estimating expected returns for each firm in the sample using a market model calibrated over a pre-event estimation window. For every security, daily returns are regressed on the S&P 500 index to obtain firm-specific alpha and beta values that characterize normal return behavior under typical market conditions. This estimation window is chosen to precede the event window so that parameter estimates are not influenced by sustainability-related political developments.

Using these estimated parameters, abnormal returns are calculated for each firm on each trading day within the event window. Abnormal returns are defined as the difference between realized returns and expected returns implied by the market

model. This step isolates firm-level deviations that cannot be explained by general market movements and therefore captures potential responses to sustainability-related information.

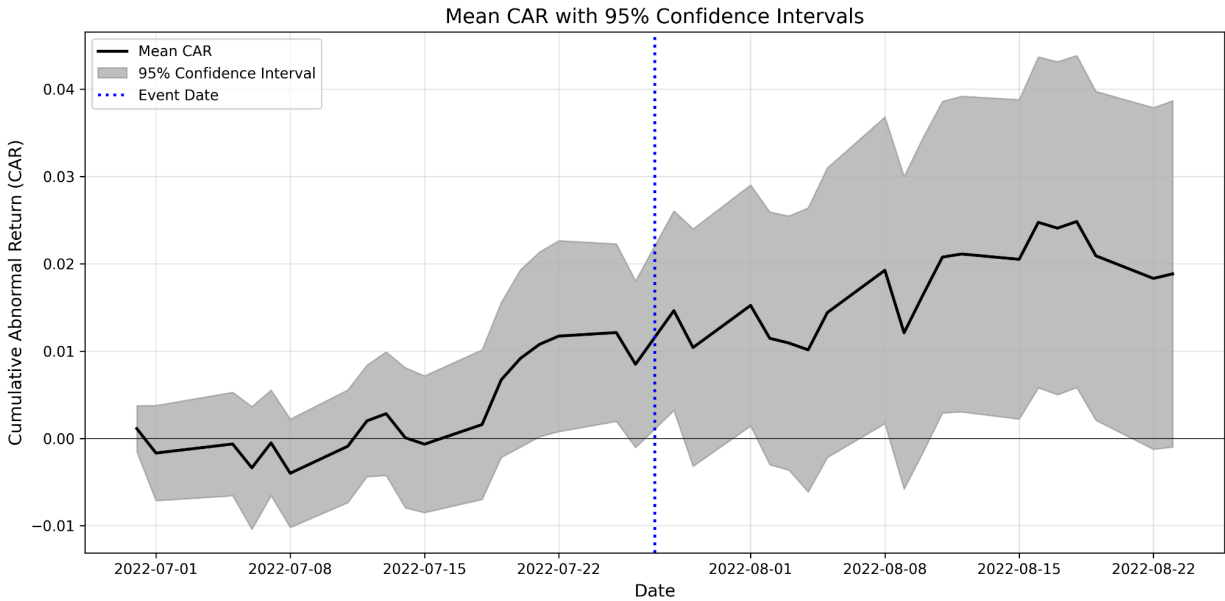
Abnormal returns are then aggregated over time to form cumulative abnormal returns for each firm. The cumulative abnormal return reflects the total market response over the event window rather than isolated daily fluctuations. This aggregation is essential for identifying persistent revaluation effects that arise when political developments alter expectations about long-run regulation, subsidies, or investment incentives.

After computing firm-level CARs, the analysis aggregates these measures across the sustainability-aligned sample. Mean cumulative abnormal returns are calculated by averaging CARs across all firms in the ESG sample for each day in the event window. This step produces a time series that reflects the collective response of sustainability-focused firms to the political event under consideration.

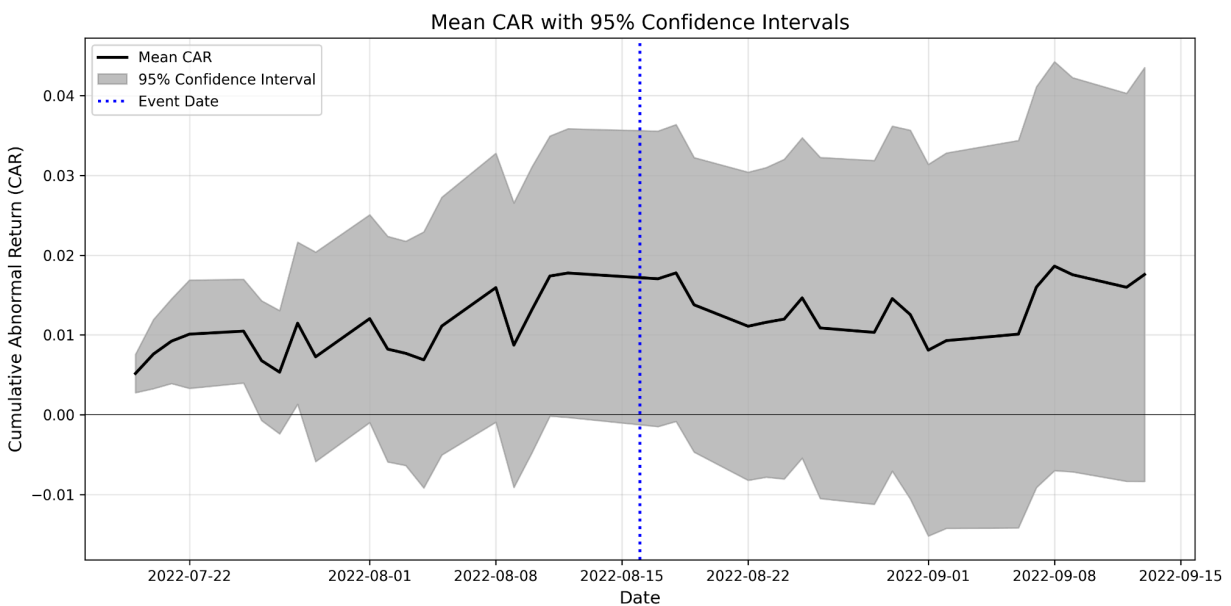
To assess the statistical reliability of these average effects, confidence intervals are constructed around the mean CAR series. These intervals account for cross-sectional variation in abnormal returns and allow the analysis to distinguish systematic market responses from noise. Periods in which confidence intervals exclude zero indicate statistically meaningful deviations from normal market behavior.

The event window is structured symmetrically around each event. This design ensures that observed post-event movements reflect market adjustment.

This full procedure is first applied to the Manchin Surprise. Following the unexpected announcement of Senator Manchin's support for climate legislation, mean cumulative abnormal returns for the ESG sample exhibit a clear upward shift in the post-event period. The timing of this change aligns closely with the event date, indicating that markets reassessed sustainability-aligned firms when legislative feasibility changed abruptly.



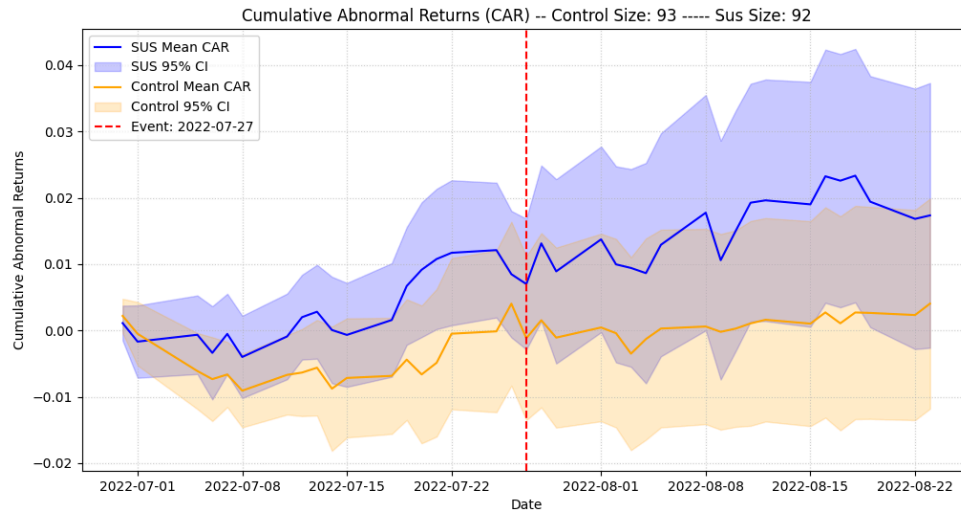
The same methodology is then applied to the Inflation Reduction Act event window. While CARs are calculated in an identical manner, the resulting pattern differs substantially. Mean cumulative abnormal returns display weaker directional movement, and confidence intervals frequently overlap with zero, suggesting that the market did not experience a sharp reassessment at the time of the IRA's formal passage.



Comparing the two sets of results reveals a pronounced contrast in market response. The Manchin Surprise generates a more distinct and persistent deviation in cumulative abnormal returns, while the IRA produces a comparatively muted and diffused pattern. This difference indicates that the magnitude of market reaction depends less on the scale of the policy and more on whether the event delivers **unexpected** information.

The contrast between the CAR trajectories supports the use of the Manchin Surprise as the primary identification event. Because the IRA's provisions were gradually revealed and debated in advance, much of its economic content appears to have been incorporated into prices prior to enactment. In contrast, the Manchin Surprise functions as a concentrated information shock, producing a clearer and more identifiable market response under the event-study framework.

With attention centered on the Manchin Surprise, the cumulative abnormal return evidence provides a clear indication that markets reacted in a differentiated manner across firm types. Sustainability-aligned firms exhibit a noticeable and sustained increase in cumulative abnormal returns following the announcement, while the matched control group remains close to zero throughout the post-event window. The separation between the two series becomes more pronounced over time, suggesting that investors revised expectations for ESG-focused firms in response to the sudden increase in the likelihood of major climate legislation. This divergence is not transient, but persists well beyond the immediate announcement period, indicating a durable re-evaluation rather than short-term trading noise.



The confidence intervals surrounding the mean CAR series reinforce this conclusion. For sustainability-aligned firms, post-event confidence bands shift upward and increasingly exclude zero, while the control group's intervals remain centered near zero with substantial overlap throughout the window. This contrast suggests that the observed post-event gains among ESG firms are not simply driven by general market volatility, but reflect a systematic response to new political information. Visually, the CAR plots alone provide strong evidence of a differential market reaction triggered by the Manchin Surprise.

OLS Regression Results						
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Dep. Variable:	CAR	R-squared:	0.013			
Model:	OLS	Adj. R-squared:	0.013			
Method:	Least Squares	F-statistic:	31.82			
Date:	Wed, 17 Dec 2025	Prob (F-statistic):	2.02e-20			
Time:	23:50:25	Log-Likelihood:	9439.7			
No. Observations:	7030	AIC:	-1.887e+04			
Df Residuals:	7026	BIC:	-1.884e+04			
Df Model:	3					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	-0.0045	0.002	-2.925	0.003	-0.008	-0.001
SUS	0.0076	0.002	3.452	0.001	0.003	0.012
POST	0.0053	0.002	2.469	0.014	0.001	0.009
POST_SUS	0.0071	0.003	2.347	0.019	0.001	0.013
=====						
Omnibus:	1109.152	Durbin-Watson:	0.114			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	17462.932			
Skew:	0.220	Prob(JB):	0.00			
Kurtosis:	10.709	Cond. No.	7.01			
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Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

The regression analysis formalizes this visual evidence by testing whether the post-event response differs statistically between the sustainability-aligned firms and the control group. The positive and statistically significant coefficient on the sustainability indicator (SUS = 0.0076, $p = 0.001$) indicates that ESG-aligned firms exhibit higher cumulative abnormal returns than control firms on average. This baseline difference is consistent with the notion that sustainability-focused firms are more sensitive to policy-related information, even outside of major event windows.

The POST coefficient (0.0053, $p = 0.014$) captures the average post-event effect across all firms, regardless of sustainability classification. Its significance indicates that the Manchin announcement was associated with a modest market-wide adjustment, likely reflecting changes in aggregate expectations related to fiscal policy, energy investment, or political stability. However, this coefficient alone does not explain the divergent CAR trajectories observed in the graphical analysis.

The key result lies in the interaction term between POST and SUS (POST_SUS = 0.0071, $p = 0.019$). This coefficient measures the additional post-event effect experienced by sustainability-aligned firms relative to the control group. Its positive and statistically significant value indicates that ESG firms experienced an incremental increase in cumulative abnormal returns following the Manchin Surprise that cannot be attributed to general market movements or pre-existing differences between firm types. Economically, this interaction effect represents the market's reassessment of sustainability-aligned firms' future profitability in light of increased policy credibility.

Although the model's R-squared is relatively small, this outcome is typical in event-study regressions involving returns data, which are inherently noisy and influenced by many unobserved factors. Importantly, the overall F-statistic is large and highly significant, indicating that the regressors jointly explain meaningful variation in cumulative abnormal returns. The statistical significance of the event-related coefficients, rather than overall explanatory power, is the relevant criterion for hypothesis testing in this setting.

Taken together, the CAR evidence and the regression results provide consistent and compelling support for the alternative hypothesis, indicating that the Manchin Surprise generated a measurable and economically meaningful market response. Intuitively, this outcome reflects how investors process political information when it directly alters the perceived feasibility of future policy rather than merely confirming an expected outcome. The Manchin announcement resolved a key source of uncertainty surrounding U.S. climate legislation by signaling that a pivotal political constraint had been lifted. For sustainability-aligned firms, this shift translated into higher expected future cash flows through anticipated subsidies, tax credits, regulatory advantages, and increased public investment, all of which are capitalized into prices immediately. The absence of a comparable response among control firms suggests that the effect was not driven by general optimism or macroeconomic sentiment, but by firm-specific exposure to climate and energy policy. In this sense, the Manchin Surprise acted as a credibility shock rather than a policy shock, triggering belief revision about long-run economic conditions facing ESG firms. This mechanism explains both the persistence of post-event cumulative abnormal returns and the statistical significance of the

interaction term in the regression, reinforcing the conclusion that sustainability-related political developments have become an identifiable and economically relevant factor in asset pricing when they unexpectedly alter policy expectations.

❖ Looking Forward

The results point to a clear consensus that the Manchin Surprise generated an observable and statistically meaningful market response, particularly among sustainability-aligned firms. The cumulative abnormal return evidence and the regression results both indicate that this political development altered investor expectations in a way that cannot be attributed to broad market movements alone. As such, the findings support the baseline hypothesis that sustainability-related political information can influence asset prices when it meaningfully changes perceptions of policy credibility.

The more substantive question that follows is not whether an effect exists, but how large that effect is in relative terms. Financial markets have reacted to numerous sustainability-related events over the past decade, including international climate agreements, domestic legislation, regulatory announcements, and green-finance initiatives, each differing in scale, surprise, and perceived durability. Applying the same event-study and regression framework to a set of comparable sustainability events would allow their impacts to be measured consistently and evaluated in relation to one another.

Extending the analysis in this way would place the Manchin Surprise within a broader empirical context and help clarify the magnitude of sustainability-driven policy shocks. Including major non-sustainability policy events, such as fiscal or industrial policy announcements, as benchmarks could further sharpen this comparison. Such an approach would help determine whether sustainability policies generate relatively modest market responses or, under conditions of heightened credibility and surprise, produce effects comparable to more traditional drivers of financial market revaluation.

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